

claude-windows / 188dd06f-9c99-487c-aa7d-b0bee83c629c

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\188dd06f-9c99-487c-aa7d-b0bee83c629c\subagents\workflows\wf_416cc0e5-a35\agent-a823b2c089c52888e.jsonl`
- SHA-256: `48d3ddc113ae5f4dfe1ab4703168c43622f32bd08305b6f20717206b3c91071d`
- Source modified: `2026-06-22T09:48:50+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_416cc0e5-a35`
- session_id: `188dd06f-9c99-487c-aa7d-b0bee83c629c`
```

Transcript

1. user (2026-06-22T09:45:35.971Z)

Review the NIGHTLY REEL-COUNT engine-regression change under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e (a NEW, additive PR). It runs the configured WASB pipeline on a GCP GPU VM nightly, asserts reel-count (==) + quality metrics + cost, emails a report, self-deletes. Files: scripts/nightly_reelcount.py, scripts/nightly_email.py, deploy/gcp/nightly_regression/run_on_vm.sh, deploy/gcp/nightly_regression/launch.sh, deploy/gcp/nightly_regression/register_scheduler.sh, deploy/gcp/nightly_regression/Dockerfile.launcher, eval_baselines/reelcount_manifest.json. Read them under that absolute path. Cite file:line. Report only REAL defects (correctness/safety/security that would bite in production or a regression that escapes), not style. Return raw structured findings.

ADVERSARIALLY VERIFY this finding ? confirm with concrete code evidence under C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/nightly-e2e or REFUTE it (default real=false if the evidence isn't clearly there). Finding:

```
{
  "title": "CRITICAL: UPTIME calculation can silently underflow to zero if BOOT_EPOCH fails to set",
  "file": "deploy/gcp/nightly_regression/run_on_vm.sh",
  "line": "15, 80",
  "severity": "high",
  "detail": "Line 15: BOOT_EPOCH=$(date +%s) ? if this fails (theoretically low probability on a fresh VM but non-zero, e.g., timezone database corrupted), BOOT_EPOCH is unset. Line 80: UPTIME=$(( $(date +%s) - BOOT_EPOCH )) ? with unset BOOT_EPOCH, bash arithmetic treats it as 0, causing UPTIME to be the full wall time, which is INCORRECT for cost billing. The Python script receives a falsely inflated uptime (the entire pipeline runs at once rather than measuring boot overhead separately). This could cause billing cost estimates to be significantly overstated or understated.",
  "fix": "Validate BOOT_EPOCH after setting: BOOT_EPOCH=$(date +%s) || { echo \"FATAL: cannot read system time\"; exit 1; }. Also add a sanity check before passing to Python: [ \"$UPTIME\" -lt 5 ] && echo \"WARN: UPTIME suspiciously low ($UPTIME s)\""}
}
```

2. assistant (2026-06-22T09:45:38.378Z)

I'll systematically analyze these files to verify the claimed defect about UPTIME calculation underflow. Let me start by reading all the relevant files.

3. user (2026-06-22T09:45:39.372Z)

```
1      #!/usr/bin/env bash
2      # Startup-script for the NIGHTLY engine-regression VM. Runs UNATTENDED on boot, then DELETES the VM
3      # (cost rule: a running GPU VM is the only real cost). Survives SSH drops (it's a startup-script, not
```

```

4      # an interactive session). Reads its config from instance metadata (set by launch.sh).
5      #
6      # Flow: Ops Agent ? pull repo @ HEAD from GCS ? build venv + WASB env ? stage
weights ?
7      #      run scripts/nightly_reelcount.py on the gs:// golden clips (emails the report)
?
8      #      upload report to GCS ? SELF-DELETE (even on failure, via the EXIT trap).
9      #
10     # Metadata keys (launch.sh sets these): repo-tgz-uri, git-sha, gcs-bucket, report-
prefix,
11     # weights-uri, email (true/false). NOT verified in CI ? dry-run via launch.sh on a
real VM first.
12     set -uo pipefail
13     exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14     echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15     BOOT_EPOCH=$(date +%s)
16
17     md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18     NAME=$(md instance/name)
19     ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20     REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21     GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22     BUCKET=$(md instance/attributes/gcs-bucket || true)
23     REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24     WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25     DO_EMAIL=$(md instance/attributes/email || echo true)
26     SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27     SMTP_USER=$(md instance/attributes/smtp-user || true)
28     EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29     DATE=$(date -u +%F)
30
31     # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32     cleanup() {
33         echo "==== self-delete $NAME ($ZONE) ====="
34         gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35             echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36     }
37     trap cleanup EXIT
38
39     fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40         echo "RUN FAILED ? attempting a failure alert"
41         if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42             cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43                 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44                 python - <<PY || true
45         from scripts.nightly_email import send_report
46         send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47             "The nightly engine-regression run failed on the VM before producing a
report.\n"
48             "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49         PY
50         fi
51     }
52     trap 'fail_email' ERR
53
54     # 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
55     curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
56         sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent
install failed (non-fatal)"

```

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57
58     # 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the
worker SA).
59     mkdir -p ~/rally && cd ~/rally
60     gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
61     tar -xzf /tmp/rally.tgz -C ~/rally
62     echo "$GIT_SHA" > ~/rally/GIT_SHA
63
64     # 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per
deploy/gcp/README §4).
65     sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
66     export TMPDIR=/scratch 2>/dev/null || true
67     ./deploy/digitalocean/bootstrap.sh --gpu
68     . .venv/bin/activate
69     pip install -e '[storage-gcs]'
70     bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned
nonzero ? continuing"
71
72     # 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
73     if [ -n "$WEIGHTS_URI" ]; then
74         mkdir -p ~/models/WASB-SBDT/pretrained_weights
75         gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar
76         export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
77     fi
78
79     # 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM
uptime so far.
80     UPTIME=$(( (date +%s) - BOOT_EPOCH ))
81     EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
82     set +e
83     NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
84     python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
85     RUN_RC=$?
86     set -e
87     echo "runner exit code: $RUN_RC (0 pass / 1 regression / 4 stale)"
88
89     # 6) Upload the dated report to GCS for the record (alongside the email).
90     if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
91         gcloud storage cp -r "output/nightly_reelcount/$DATE."*
"gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
92         gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/"
|| true
93     fi
94
95     echo "==== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
96     trap - ERR # disable the failure-email trap; the EXIT (self-delete) trap stays
97     # the EXIT trap deletes the VM
98

```

4. user (2026-06-22T09:45:40.543Z)

```

1     ""Nightly END-TO-END engine regression ? reel-count + quality + cost (the GCP-VM
nightly).
2
3     Runs the **configured + checkpointed** pipeline (the real WASB GPU detector, as a
deployment serves
4     it) end-to-end on 2-3 short owned golden videos and asserts the **number of reels does
not change**
5     vs a frozen baseline. Unlike ``scripts/nightly_regression.py`` (PR #202 ? a CPU re-score
of CACHED
6     trajectories, the windowing/scorer only), this one runs the *full* pipeline on the *real
model*, so it
7     needs a GPU ? hence it runs on an ephemeral GCP GPU VM

```

```

(``deploy/gcp/nightly_regression/``), emails the
8     owner the report, and self-deletes. It is **default-additive**: a tripwire, not the
promotion gate.
9
10    What it asserts / reports, per the owner's ask ("quality metrics + regression report +
cost"):
11    ? REEL COUNT per video ? ``len(play_segments)`` after segmentation; the detected
highlight clips.
12    Pure-CV / WASB detection is deterministic for a fixed (video, config, weights), so
this is asserted
13    **EXACT** (with a re-run-once guard for the one known flake: a transient DB-insert-
None, see $guard).
14    ? QUALITY METRICS per video (when golden labels are provided) ? the R2 tolerance-match
block from
15    ``backend.eval.rally_seg_eval`` (tol_f1 / f1@0.5 / f1@0.25), compared with a small
tolerance.
16    ? COST of the run ? VM uptime (or processing wall) × machine $/hr via
``backend.billing`` ? USD + INR.
17
18    Extensible pattern (add a video = one manifest row + re-freeze):
19    ``eval_baselines/reelcount_manifest.json`` ? the videos (gs:// / local / gdrive://
URIs) + segmenter
20    ``eval_baselines/reelcount_baseline.json`` ? the frozen expected counts (regenerated,
never hand-edited)
21
22    Usage
23    -----
24    Freeze the baseline from the current configured+checkpointed engine (after an intended
change, or first run):
25    python scripts/nightly_reelcount.py --update-baseline
26
27    Nightly check (exit 1 on regression; writes output/nightly_reelcount/<date>.md; emails
if --email):
28    python scripts/nightly_reelcount.py --email
29
30    Exit codes: 0 pass · 1 regression (reel-count changed / a video errored / quality
dropped) · 2 operational
31    (manifest/videos missing, nothing ran) · 3 no baseline · 4 stale (corpus/segmenter
changed ? re-freeze).
32    """
33    from __future__ import annotations
34
35    import argparse
36    import datetime as _dt
37    import json
38    import os
39    import sys
40    from dataclasses import dataclass, field
41    from typing import Any, Dict, List, Optional, Tuple
42
43    REPO_ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
44    if REPO_ROOT not in sys.path:
45        sys.path.insert(0, REPO_ROOT)
46
47    # billing is pure (typing only) ? safe to import at module top.
48    from backend import billing # noqa: E402
49
50    DEFAULT_MANIFEST = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_manifest.json")
51    DEFAULT_BASELINE = os.path.join(REPO_ROOT, "eval_baselines", "reelcount_baseline.json")
52    DEFAULT_REPORT_DIR = os.path.join(REPO_ROOT, "output", "nightly_reelcount")
53
54    #: Quality metrics we report + (when labels exist) gate ? higher is better, small
tolerance.
55    QUALITY_HIGHER: Tuple[str, ...] = ("tol_f1", "f1@0.5", "f1@0.25")
56    DEFAULT_QUALITY_TOL = 0.02 # quality is float/labeled ? a looser tripwire than the

```

```

exact reel-count
57     DEFAULT_FX_INR_PER_USD = 83.0
58     #: Fallback machine price if the manifest doesn't pin one (T4 on n1-standard-8, asia-
south1, ~mid-2026).
59     DEFAULT_MACHINE_USD_PER_HR = 0.35
60
61
62     @dataclass(frozen=True)
63     class Tolerances:
64         quality_tol: float = DEFAULT_QUALITY_TOL
65
66         def to_dict(self) -> Dict[str, Any]:
67             return {"quality_tol": self.quality_tol}
68
69
70     # ----- #
71     # Pure cost math (wraps backend.billing ? no IO).
72     # ----- #
73     def cost_report(billed_seconds: float, machine_usd_per_hr: float,
74                     fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
75         """Run cost from billed wall-seconds × machine rate ? USD + INR (``backend.billing``
arithmetic).
76
77         ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes
``--vm-uptime-seconds``;
78         otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays
for boot/install)."""
79         rate_per_sec = float(machine_usd_per_hr) / 3600.0
80         usage = {billing.WALL_SECONDS: float(billed_seconds)}
81         rates = {billing.WALL_SECONDS: rate_per_sec}
82         usd, breakdown = billing.compute_cost_usd(usage, rates)
83         return {
84             "billed_seconds": round(float(billed_seconds), 1),
85             "gpu_seconds": round(float(gpu_seconds), 1),
86             "machine_usd_per_hr": float(machine_usd_per_hr),
87             "usd": usd,
88             "inr": billing.to_inr(usd, fx_inr_per_usd),
89             "breakdown": breakdown,
90         }
91
92
93     # ----- #
94     # Pure summary + comparison core (no IO ? unit-testable with plain dicts).
95     # ----- #
96     def summarize_results(run_results: List[Dict[str, Any]], segmenter: str,
97                           cost: Dict[str, Any]) -> Dict[str, Any]:
98         """Compact, comparable snapshot from per-video run results.
99
100         Each run result: ``{name, reel_count, ok, error, quality: {tol_f1,...}}None,
wall_seconds}``.
101         A video that errored keeps ``reel_count=None`` + ``ok=False`` (so the diff flags it,
never hides it)."""
102         per_video: Dict[str, Any] = {}
103         for r in run_results:
104             per_video[r["name"]] = {
105                 "reel_count": r.get("reel_count"),
106                 "ok": bool(r.get("ok", False)),
107                 "error": r.get("error"),
108                 "quality": r.get("quality"),
109             }
110         return {"segmenter": segmenter, "cost": cost, "per_video": per_video,
111               "n": len(per_video)}
112
113
114     @dataclass

```

```

115     class Finding:
116         video: str
117         metric: str                # "reel_count" | "error" | a quality key | "segmenter" |
"corpus"
118         kind: str                  # "regression" | "improvement" | "stale"
119         baseline: Optional[float] = None
120         current: Optional[float] = None
121         detail: str = ""
122
123     def message(self) -> str:
124         if self.kind == "stale":
125             return f"[STALE] {self.metric}: {self.detail}"
126         if self.metric == "error":
127             return f"[REGRESSION] {self.video}: pipeline ERROR ? {self.detail}"
128         b = "?" if self.baseline is None else f"{self.baseline:g}"
129         c = "?" if self.current is None else f"{self.current:g}"
130         tag = "REGRESSION" if self.kind == "regression" else "improvement"
131         return f"[{tag}] {self.video}/{self.metric}: {b} ? {c}"
132
133
134     @dataclass
135     class NightlyResult:
136         findings: List[Finding] = field(default_factory=list)
137         added: List[str] = field(default_factory=list)
138         removed: List[str] = field(default_factory=list)
139
140         @property
141         def regressions(self) -> List[Finding]:
142             return [f for f in self.findings if f.kind == "regression"]
143
144         @property
145         def improvements(self) -> List[Finding]:
146             return [f for f in self.findings if f.kind == "improvement"]
147
148         @property
149         def stale(self) -> List[Finding]:
150             return [f for f in self.findings if f.kind == "stale"]
151
152         def overall_status(self) -> str:
153             if self.regressions:
154                 return "REGRESSION"
155             if self.stale:
156                 return "STALE"
157             return "PASS"
158
159         def exit_code(self) -> int:
160             if self.regressions:
161                 return 1
162             if self.stale:
163                 return 4
164             return 0
165
166
167     def compare(baseline: Dict[str, Any], current: Dict[str, Any], tols: Tolerances) ->
NightlyResult:
168         """Diff current run vs frozen baseline.
169
170         * Segmenter changed ? STALE (numbers not comparable; re-freeze).
171         * Corpus (video set) changed ? STALE + report added/removed; still diff the
INTERSECTION.
172         * Per shared video: reel_count EXACT (any change = regression); a pipeline ERROR =
regression;
173         quality metrics drop beyond ``quality_tol`` = regression (rise = improvement).
174         """
175         res = NightlyResult()

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```

176         if baseline.get("segmenter") != current.get("segmenter"):
177             res.findings.append(Finding("", "segmenter", "stale",
178                                     detail=f"{baseline.get('segmenter')} ?
{current.get('segmenter')}}"))
179         return res
180
181     b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
182     res.added = sorted(set(c_pv) - set(b_pv))
183     res.removed = sorted(set(b_pv) - set(c_pv))
184     if res.added or res.removed:
185         res.findings.append(Finding("", "corpus", "stale",
186                                     detail=f"added={res.added} removed={res.removed}"))
187
188     for v in sorted(set(b_pv) & set(c_pv)):
189         b, c = b_pv[v], c_pv[v]
190         # 1) pipeline must still succeed
191         if not c.get("ok", False) or c.get("reel_count") is None:
192             res.findings.append(Finding(v, "error", "regression",
193                                     detail=str(c.get("error") or "no reel_count
produced")))
194             continue
195         # 2) reel count ? EXACT
196         bc, cc = b.get("reel_count"), c.get("reel_count")
197         if bc is not None and cc is not None and cc != bc:
198             res.findings.append(Finding(v, "reel_count", "regression", float(bc),
float(cc)))
199         # 3) quality metrics ? tolerance (only if both sides have them)
200         bq, cq = b.get("quality") or {}, c.get("quality") or {}
201         for m in QUALITY_HIGHER:
202             bv, cv = bq.get(m), cq.get(m)
203             if bv is None or cv is None:
204                 continue
205             if cv < bv - tols.quality_tol:
206                 res.findings.append(Finding(v, m, "regression", bv, cv))
207             elif cv > bv + tols.quality_tol:
208                 res.findings.append(Finding(v, m, "improvement", bv, cv))
209     return res
210
211
212     # ----- #
213     # Report formatting (pure).
214     # ----- #
215     def _q(qd: Optional[Dict[str, Any]], key: str) -> str:
216         if not qd or qd.get(key) is None:
217             return "?"
218         return f"{qd[key]:.3f}"
219
220
221     def format_report(baseline: Dict[str, Any], current: Dict[str, Any],
222                      result: NightlyResult, date: str, head: str) -> str:
223         status = result.overall_status()
224         badge = {"PASS": "? PASS", "REGRESSION": "? REGRESSION",
225                 "STALE": "? STALE (re-freeze the baseline)"}[status]
226         cost = current.get("cost", {})
227         L: List[str] = [
228             f"# Nightly engine regression (reel-count + quality + cost) ? {date}",
229             "",
230             f"**Status: {badge}** · exit code {result.exit_code()} · segmenter
`{current.get('segmenter')}`",
231             "",
232             f"- HEAD `{head}` · baseline `{baseline.get('git_commit', '?')}` (frozen
{baseline.get('generated_at', '?')})",
233             f"- **Run cost: ${cost.get('usd', 0):.4f} (?{cost.get('inr', 0):.2f})* · "
234             f"billed {cost.get('billed_seconds', 0):.0f}s @ ${cost.get('machine_usd_per_hr',
0)}/hr",

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235         "",
236     ]
237     if result.regressions:
238         L += ["## ? Regressions", ""] + [f"- {f.message()}" for f in result.regressions]
239     + [""]
240     if result.stale:
241         L += ["## ? Stale / not comparable", ""] + [f"- {f.message()}" for f in
result.stale]
242
243         L += ["- Action: `python scripts/nightly_reelcount.py --update-baseline` (after
confirming the change is intended).", ""]
244
245         # Per-video table.
246         b_pv, c_pv = baseline.get("per_video", {}), current.get("per_video", {})
247         L += ["## Per-video", ""]
248         "| video | reels (base?now) | tol_f1 (base?now) | f1@0.5 | status |",
249         "|---|---|---|---|---|"
250         for v in sorted(set(b_pv) | set(c_pv)):
251             b, c = b_pv.get(v, {}), c_pv.get(v, {})
252             bc = b.get("reel_count")
253             cc = c.get("reel_count")
254             rc = f"'?' if bc is None else bc?{'ERROR' if not c.get('ok', True) else ('?'
if cc is None else cc)}"
255             bq, cq = b.get("quality"), c.get("quality")
256             tf = f"_q(bq, 'tol_f1')?_q(cq, 'tol_f1')?"
257             f5 = f"_q(bq, 'f1@0.5')?_q(cq, 'f1@0.5')?"
258             vstatus = "?" if any(f.video == v and f.kind == "regression" for f in
result.findings) else "?"
259             L.append(f"| {v} | {rc} | {tf} | {f5} | {vstatus} |")
260         L.append("")
261
262         if result.improvements:
263             L += ["_Improvements over baseline (re-freeze to ratchet up):_"] \
+ [f"- {f.message()}" for f in result.improvements] + [""]
264         L += ["---",
"_Full configured+checkpointed pipeline on the real model (GCP GPU VM).
Tripwire only ? does "
265             "not change the promotion gate. Complements PR #202's cached-trajectory CPU
re-score._"]
266         return "\n".join(L)
267
268
269     # ----- #
270     # IO shell ? running the configured pipeline per video (needs backend + GPU + video).
271     # ----- #
272     def _load_gts(labels_ref: Optional[str]) -> Optional[List[Tuple[float, float]]]:
273         """Load ground-truth rally intervals from a golden-features JSON (`gts` key) or a
rallies CSV
274         (`start,end` columns / pairs). Returns None when no labels are configured."""
275         if not labels_ref:
276             return None
277         from backend.storage import fetch
278         local = fetch(labels_ref)
279         if local.endswith(".json"):
280             with open(local, encoding="utf-8") as fh:
281                 data = json.load(fh)
282                 return [(float(s), float(e)) for s, e in (data.get("gts") or [])]
283         # CSV: header start,end (or first two numeric columns)
284         import csv
285         out: List[Tuple[float, float]] = []
286         with open(local, encoding="utf-8") as fh:
287             for row in csv.reader(fh):
288                 try:
289                     out.append((float(row[0]), float(row[1])))
290                 except (ValueError, IndexError):
291                     continue

```



```

292         return out
293
294
295     def _segments_to_intervals(segs: List[Dict[str, Any]]) -> List[Tuple[float, float]]:
296         """(start, end) pairs from segmenter result dicts, tolerant of the two key
conventions in the
297         codebase (``start_time``/``end_time`` ? motion/DB shape ? or ``start``/``end``).
Used only for the
298         optional quality metrics; the reel COUNT is ``len(segs)`` regardless, so a key
mismatch degrades
299         quality scoring gracefully (skipped) without ever miscounting reels."""
300         out: List[Tuple[float, float]] = []
301         for s in segs:
302             start = s.get("start_time", s.get("start"))
303             end = s.get("end_time", s.get("end"))
304             if start is not None and end is not None:
305                 out.append((float(start), float(end)))
306         return out
307
308
309     def run_one_video(video: Dict[str, Any], segmenter: str, config: Any,
310                      rerun_on_mismatch: bool = True,
311                      baseline_count: Optional[int] = None) -> Dict[str, Any]:
312         """Run the configured pipeline on one video ? reel count (+ quality vs labels). Lazy
backend imports.
313
314         The re-run-once guard (the one known non-determinism ? a transient
``add_play_segment``?None drop,
315         database.py): if the count != the frozen baseline, re-process the video ONCE; a
transient drop won't
316         reproduce, a real change will."""
317         import random
318         import time
319
320         from backend.eval import rally_seg_eval
321         from backend.infrastructure.database import Database
322         from backend.pipeline.segmenters import segmenter_registry
323         from backend.results import Failure
324         from backend.storage import fetch
325         from backend.utils.hashing import compute_video_id
326
327         name = video["name"]
328         try:
329             local = fetch(video["uri"])
330         except Exception as e: # noqa: BLE001 ? surface, never hide
331             return {"name": name, "reel_count": None, "ok": False, "error": f"fetch failed:
{e}",
332                    "quality": None, "wall_seconds": 0.0}
333
334         gts = _load_gts(video.get("labels_uri"))
335         seg_cls = segmenter_registry.get(segmenter)
336         if not seg_cls:
337             return {"name": name, "reel_count": None, "ok": False,
338                    "error": f"segmenter '{segmenter}' not registered", "quality": None,
"wall_seconds": 0.0}
339
340         def _process(db_path: str) -> Tuple[Optional[int], Optional[List[Tuple[float,
float]]], float, Optional[str]]:
341             random.seed(0) # close motion.py's unseeded metadata RNG so secondary fields
are reproducible too
342             db = Database(db_path)
343             video_id = compute_video_id(local)
344             t0 = time.monotonic()
345             result = seg_cls(db, config).process_video(video_id, local)
346             wall = time.monotonic() - t0

```

```

347         if isinstance(result, Failure):
348             return None, None, wall, getattr(result, "message", "segmenter failed")
349         segs = result.value if hasattr(result, "value") else result
350         intervals = _segments_to_intervals(segs)
351         return len(segs), intervals, wall, None
352
353     import tempfile
354     total_wall = 0.0
355     with tempfile.TemporaryDirectory(prefix="nightly_reel_") as td:
356         count, intervals, wall, err = _process(os.path.join(td, "run1.db"))
357         total_wall += wall
358         if err is not None:
359             return {"name": name, "reel_count": None, "ok": False, "error": err,
360                    "quality": None, "wall_seconds": round(total_wall, 2)}
361         # re-run-once guard for the transient DB-drop flake
362         if rerun_on_mismatch and baseline_count is not None and count != baseline_count:
363             count2, intervals2, wall2, err2 = _process(os.path.join(td, "run2.db"))
364             total_wall += wall2
365             if err2 is None and count2 is not None:
366                 count, intervals = count2, intervals2 # the reproduced (or corrected)
value
367
368         quality = None
369         if gts and intervals is not None:
370             row = rally_seg_eval.score_intervals(intervals, gts)
371             quality = {k: row[k] for k in ("tol_f1", "f1@0.5", "f1@0.25",
"tol_precision", "tol_recall")}
372                 if k in row}
373
374         return {"name": name, "reel_count": count, "ok": True, "error": None,
375                "quality": quality, "wall_seconds": round(total_wall, 2)}
376
377
378     def load_manifest(path: str) -> Dict[str, Any]:
379         with open(path, encoding="utf-8") as fh:
380             return json.load(fh)
381
382
383     def run_corpus(manifest: Dict[str, Any], baseline: Optional[Dict[str, Any]],
384                   rerun_on_mismatch: bool = True) -> Tuple[List[Dict[str, Any]], float]:
385         """Run every manifest video through the configured pipeline. Returns (run_results,
total_wall_s)."""
386         from backend.config import load_config
387
388         config = load_config()
389         # apply manifest config overrides onto the typed config (dotted keys)
390         for dotted, val in (manifest.get("config_overrides") or {}).items():
391             _set_dotted(config, dotted, val)
392         segmenter = manifest.get("segmenter") or getattr(getattr(config, "indexing", None),
"default_segmenter", "motion")
393         b_pv = (baseline or {}).get("per_video", {})
394         results: List[Dict[str, Any]] = []
395         total_wall = 0.0
396         for v in manifest.get("videos", []):
397             base_count = (b_pv.get(v["name"]) or {}).get("reel_count")
398             r = run_one_video(v, segmenter, config, rerun_on_mismatch, base_count)
399             total_wall += float(r.get("wall_seconds") or 0.0)
400             results.append(r)
401         return results, total_wall
402
403
404     def _set_dotted(config: Any, dotted: str, value: Any) -> None:
405         """Best-effort dotted-path set onto a pydantic config (e.g.
'indexing.motion_threshold')."""
406         parts = dotted.split(".")

```

```

407     obj = config
408     for p in parts[:-1]:
409         obj = getattr(obj, p, None)
410         if obj is None:
411             return
412     try:
413         setattr(obj, parts[-1], value)
414     except Exception: # noqa: BLE001 ? overrides are best-effort; a bad key shouldn't
crash the run
415         pass
416
417
418     def _git_head() -> str:
419         import subprocess
420         try:
421             out = subprocess.run(["git", "-C", REPO_ROOT, "rev-parse", "--short", "HEAD"],
422                                 capture_output=True, text=True, timeout=10)
423             return out.stdout.strip() or "?"
424         except Exception: # noqa: BLE001
425             return "?"
426
427
428     def load_baseline(path: str) -> Optional[Dict[str, Any]]:
429         if not os.path.exists(path):
430             return None
431         with open(path, encoding="utf-8") as fh:
432             return json.load(fh)
433
434
435     def save_baseline(path: str, snapshot: Dict[str, Any]) -> None:
436         out = dict(snapshot)
437         out["generated_at"] = _dt.date.today().isoformat()
438         out["git_commit"] = _git_head()
439         out["_doc"] = ("Frozen nightly reel-count baseline. Regenerate with `python
scripts/nightly_reelcount.py "
440                       "--update-baseline` after an INTENDED engine change or when adding a
video. Tied to the "
441                       "configured+checkpointed engine + the manifest corpus.")
442         out.pop("cost", None) # cost is per-run, not part of the frozen baseline
443         os.makedirs(os.path.dirname(path), exist_ok=True)
444         tmp = path + ".tmp"
445         with open(tmp, "w", encoding="utf-8") as fh:
446             json.dump(out, fh, indent=2, sort_keys=True)
447             fh.write("\n")
448         os.replace(tmp, path)
449
450
451     def build_parser() -> argparse.ArgumentParser:
452         p = argparse.ArgumentParser(description="Nightly E2E engine regression: reel-count +
quality + cost.")
453         p.add_argument("--manifest", default=DEFAULT_MANIFEST)
454         p.add_argument("--baseline", default=DEFAULT_BASELINE)
455         p.add_argument("--report-dir", default=DEFAULT_REPORT_DIR)
456         p.add_argument("--quality-tol", type=float, default=DEFAULT_QUALITY_TOL)
457         p.add_argument("--vm-uptime-seconds", type=float, default=None,
458                       help="authoritative VM uptime (boot?delete) for the cost line; "
459                       "defaults to the in-process pipeline wall time (a lower bound)")
460         p.add_argument("--no-rerun-guard", action="store_true",
461                       help="disable the re-run-once-on-mismatch guard (the DB-drop flake
mitigation)")
462         p.add_argument("--update-baseline", action="store_true",
463                       help="freeze the current reel counts + quality as the new baseline
instead of checking")
464         p.add_argument("--email", action="store_true", help="email the report (see
scripts/nightly_email.py)")

```

```

465     p.add_argument("--email-to", default=None, help="override the report recipient")
466     p.add_argument("--date", default=None, help="override the report date (YYYY-MM-DD;
for tests)")
467     p.add_argument("--no-report", action="store_true", help="skip writing the dated
report file")
468     return p
469
470
471     def main(argv: Optional[List[str]] = None) -> int:
472         import logging
473         logging.basicConfig(level=logging.INFO, format="%(levelname)s %(message)s")
474         log = logging.getLogger("nightly_reelcount")
475         args = build_parser().parse_args(argv)
476         tols = Tolerances(quality_tol=args.quality_tol)
477         date = args.date or _dt.date.today().isoformat()
478
479         if not os.path.exists(args.manifest):
480             log.error("manifest not found: %s", args.manifest)
481             return 2
482         manifest = load_manifest(args.manifest)
483         baseline = load_baseline(args.baseline)
484
485         try:
486             run_results, total_wall = run_corpus(manifest, baseline, rerun_on_mismatch=not
args.no_rerun_guard)
487         except Exception as e: # noqa: BLE001 ? operational failure (deps/config) ? exit 2,
never a false "pass"
488             log.error("run failed: %s", e)
489             return 2
490         if not run_results:
491             log.error("no videos in manifest %s", args.manifest)
492             return 2
493
494         billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else
total_wall
495         cost = cost_report(
496             billed_seconds=billed,
497             machine_usd_per_hr=float(manifest.get("machine_usd_per_hr",
DEFAULT_MACHINE_USD_PER_HR)),
498             fx_inr_per_usd=float(manifest.get("fx_inr_per_usd", DEFAULT_FX_INR_PER_USD)),
499             gpu_seconds=total_wall,
500         )
501         segmenter = manifest.get("segmenter", "motion")
502         snapshot = summarize_results(run_results, segmenter, cost)
503         snapshot["git_commit"] = _git_head()
504         snapshot["generated_at"] = date
505
506         if args.update_baseline:
507             save_baseline(args.baseline, snapshot)
508             log.info("baseline frozen ? %s", args.baseline)
509             for r in run_results:
510                 log.info(" %s: reels=%s ok=%s", r["name"], r.get("reel_count"),
r.get("ok"))
511             return 0
512
513         if baseline is None:
514             log.error("no baseline at %s ? freeze one with --update-baseline (on the
configured engine).", args.baseline)
515             return 3
516
517         result = compare(baseline, snapshot, tols)
518         report = format_report(baseline, snapshot, result, date, snapshot["git_commit"])
519
520         if not args.no_report:
521             os.makedirs(args.report_dir, exist_ok=True)

```

```

522         md_path = os.path.join(args.report_dir, f"{date}.md")
523         with open(md_path, "w", encoding="utf-8") as fh:
524             fh.write(report + "\n")
525         with open(os.path.join(args.report_dir, f"{date}.json"), "w", encoding="utf-8")
as fh:
526             json.dump({"date": date, "status": result.overall_status(), "exit_code":
result.exit_code(),
527                       "snapshot": snapshot, "regressions": [f.message() for f in
result.regressions]},
528                       fh, indent=2, sort_keys=True)
529         log.info("report ? %s", md_path)
530
531     print(report)
532
533     if args.email:
534         try:
535             from scripts.nightly_email import send_report
536             subject = f"[KhelSutra nightly] {result.overall_status()} ? reel-count
engine regression {date}"
537             sent = send_report(subject, report, to=args.email_to)
538             log.info("email %s", "sent" if sent else "skipped (no creds ? report still
written)")
539         except Exception as e: # noqa: BLE001 ? a broken mailer must NOT mask the
regression verdict
540             log.error("email failed (report still written): %s", e)
541
542     code = result.exit_code()
543     if code == 1:
544         log.error("REEL-COUNT REGRESSION (%d) ? see %s", len(result.regressions),
args.report_dir)
545     elif code == 4:
546         log.warning("baseline STALE ? corpus/segmenter changed; re-freeze with --update-
baseline")
547     else:
548         log.info("nightly reel-count check PASSED (cost $%.4f / ?%.2f)", cost["usd"],
cost["inr"])
549     return code
550
551
552 if __name__ == "__main__":
553     raise SystemExit(main())
554

```

5. user (2026-06-22T09:45:40.722Z)

```

1     """Email the nightly regression report ? stdlib SMTP, creds from env or GCP Secret
Manager.
2
3     Used by ``scripts/nightly_reelcount.py --email`` (and reusable by any nightly).
Deliberately
4     dependency-light (stdlib ``smtplib`` + ``email.message``) and **graceful**: if no SMTP
password can
5     be resolved it logs + returns ``False`` (the report is still written/printed ? a broken
mailer must
6     never mask a regression).
7
8     Credential resolution (first hit wins), so it works on a GCP VM, locally, or in tests:
9     ? Recipient: ``--email-to`` arg ? ``NIGHTLY_EMAIL_TO`` ? ``avi.dullu@gmail.com``
10    ? SMTP host: ``NIGHTLY_SMTP_HOST`` (default ``smtp.gmail.com``) . port
``NIGHTLY_SMTP_PORT`` (587, STARTTLS)
11    ? SMTP user: ``NIGHTLY_SMTP_USER`` (default = recipient)
12    ? SMTP pass: ``NIGHTLY_SMTP_PASS`` ? Secret Manager ``NIGHTLY_SMTP_SECRET``
13    (``projects/<p>/secrets/<name>/versions/latest``; via the google-cloud-
secret-manager
14
lib if installed, else the ``gcloud`` CLI). A Gmail **app password** is

```

```

the simplest.
15
16     Set the secret once (owner):  echo -n '<app-password>' | gcloud secrets create nightly-
smtp-pass --data-file=-
17     and on the VM:  export NIGHTLY_SMTP_SECRET=projects/<proj>/secrets/nightly-smtp-
pass/versions/latest
18                     export NIGHTLY_SMTP_USER=<your-gmail>
NIGHTLY_EMAIL_TO=avi.dullu@gmail.com
19     """
20     from __future__ import annotations
21
22     import logging
23     import os
24     from email.message import EmailMessage
25     from typing import Optional
26
27     log = logging.getLogger("nightly_email")
28
29     DEFAULT_TO = "avi.dullu@gmail.com"
30     DEFAULT_HOST = "smtp.gmail.com"
31     DEFAULT_PORT = 587
32
33
34     def build_message(subject: str, body_md: str, from_addr: str, to_addr: str) ->
EmailMessage:
35         """A text/plain email carrying the markdown report (renders fine in any client).
Pure ? testable."""
36         msg = EmailMessage()
37         msg["Subject"] = subject
38         msg["From"] = from_addr
39         msg["To"] = to_addr
40         msg.set_content(body_md)
41         return msg
42
43
44     def resolve_secret(env: Optional[dict] = None) -> Optional[str]:
45         """Resolve the SMTP password from env or GCP Secret Manager. Returns None if
unresolvable."""
46         env = env if env is not None else os.environ
47         direct = env.get("NIGHTLY_SMTP_PASS")
48         if direct:
49             return direct
50         secret_ref = env.get("NIGHTLY_SMTP_SECRET")
51         if not secret_ref:
52             return None
53         # Prefer the client lib; fall back to the gcloud CLI (no extra dependency required).
54         try:
55             from google.cloud import secretmanager # type: ignore
56             client = secretmanager.SecretManagerServiceClient()
57             return
client.access_secret_version(name=secret_ref).payload.data.decode("utf-8")
58         except Exception: # noqa: BLE001 ? lib absent or auth issue ? try the CLI
59             pass
60         try:
61             import subprocess
62             name = secret_ref.split("/secrets/", 1)[1].split("/", 1)[0] if "/secrets/" in
secret_ref else secret_ref
63             out = subprocess.run(["gcloud", "secrets", "versions", "access", "latest", "--
secret", name],
64                                 capture_output=True, text=True, timeout=30)
65             return out.stdout.strip() if out.returncode == 0 and out.stdout.strip() else
None
66         except Exception as e: # noqa: BLE001
67             log.warning("secret-manager resolution failed: %s", e)
68             return None

```

```

69
70
71     def send_report(subject: str, body_md: str, to: Optional[str] = None,
72                     env: Optional[dict] = None) -> bool:
73         """Send the report. Returns True if sent, False if skipped (no creds) ? never raises
on missing creds."""
74         env = env if env is not None else os.environ
75         to_addr = to or env.get("NIGHTLY_EMAIL_TO") or DEFAULT_TO
76         host = env.get("NIGHTLY_SMTP_HOST", DEFAULT_HOST)
77         port = int(env.get("NIGHTLY_SMTP_PORT", DEFAULT_PORT))
78         user = env.get("NIGHTLY_SMTP_USER") or to_addr
79         password = resolve_secret(env)
80         if not password:
81             log.warning("no SMTP password (set NIGHTLY_SMTP_PASS or NIGHTLY_SMTP_SECRET) ?
email skipped.")
82             return False
83
84         import smtplib
85         msg = build_message(subject, body_md, from_addr=user, to_addr=to_addr)
86         with smtplib.SMTP(host, port, timeout=60) as s:
87             s.starttls()
88             s.login(user, password)
89             s.send_message(msg)
90         log.info("emailed report to %s", to_addr)
91         return True
92

```

6. user (2026-06-22T09:45:41.803Z)

```

1     #!/usr/bin/env bash
2     # Launch ONE nightly engine-regression run on an ephemeral GCP GPU VM, then return. The
VM runs
3     # run_on_vm.sh as its startup-script (regression ? email ? SELF-DELETE), so this just
stages the repo
4     # and kicks the VM off. Driver-agnostic: call it from Cloud Scheduler?Cloud Run job, a
cron on any
5     # always-on box, the owner's machine, or a Claude routine. Needs gcloud auth + the repo
checkout.
6     #
7     #   bash deploy/gcp/nightly_regression/launch.sh           # L4, stages HEAD,
kicks the VM
8     #   RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash
deploy/gcp/nightly_regression/launch.sh
9     #
10    # ? Run this MANUALLY once as a dry-run (watch the serial log) before wiring a scheduler
? the VM-side
11    # flow is NOT covered by CI. Tail: gcloud compute instances get-serial-port-output
<name> --zone=<z>.
12    set -uo pipefail
13
14    PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
15    BUCKET="${RALLY_BUCKET:-khelesutra-rally-corpus}"
16    SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
17    GPU="${RALLY_GPU:-l4}"
18    NAME="${RALLY_VM_NAME:-rally-nightly-$(date -u +%Y%m%d)}"
19    REF="${RALLY_REF:-HEAD}"
20    WEIGHTS_URI="${RALLY_WEIGHTS_URI:-gs://$BUCKET/weights/wasb_badminton_best.pth.tar}"
21    REPORT_PREFIX="${RALLY_REPORT_PREFIX:-nightly/reports}"
22    DO_EMAIL="${RALLY_EMAIL:-true}"
23    SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-
pass/versions/latest}"
24    SMTP_USER="${NIGHTLY_SMTP_USER:-}"
25    EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
26    HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
27    STARTUP="$HERE/run_on_vm.sh"

```

```

28     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
29
30     ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a
asia-southeast1-b asia-southeast1-c}"
31     if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else
MACHINE="n1-standard-8"; ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi
32
33     # 1) Stage the repo @ REF to GCS (no GitHub auth on the VM; the worker SA can read the
bucket).
34     SHA="$(git rev-parse "$REF")"
35     TGZ_URI="gs://$BUCKET/nightly/repo/${SHA}.tgz"
36     echo "== staging repo $SHA -> $TGZ_URI =="
37     TMP_TGZ="$(mktemp --suffix=.tgz)"
38     git archive --format=tar.gz -o "$TMP_TGZ" "$REF"
39     gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
40     rm -f "$TMP_TGZ"
41
42     # 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config
metadata.
43     ERR="$(mktemp)"
44     for Z in $ZONES; do
45         echo "== creating $NAME ($GPU) in $Z =="
46         if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
47             --machine-type="$MACHINE" "${ACCEL[@]}" \
48             --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
49             --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-
platform-release \
50             --boot-disk-size=120GB --boot-disk-type=pd-balanced \
51             --service-account="$SA" --scopes=cloud-platform \
52             --metadata-from-file=startup-script="$STARTUP" \
53             --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",gcs-bucket="$BUCKET",report-
prefix="$REPORT_PREFIX",weights-uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-
secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-to="$EMAIL_TO" \
54             2>"$ERR"; then
55             echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and
self-delete."
56             echo "    serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z
--project=$PROJECT"
57             echo "    if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z
-q"
58             rm -f "$ERR"; exit 0
59         fi
60         if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo "    (no $GPU capacity in
$Z ? next zone)"; continue; fi
61         echo "    create failed: "; tail -4 "$ERR"; rm -f "$ERR"; exit 1
62     done
63     echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
64     rm -f "$ERR"; exit 1
65

```

7. user (2026-06-22T09:45:42.280Z)

```

1     {
2         "_doc": "Nightly E2E engine reel-count manifest (scripts/nightly_reelcount.py). The
configured+checkpointed pipeline runs on each video on a GCP GPU VM; the reel count + quality
metrics are asserted vs eval_baselines/reelcount_baseline.json. ADD A VIDEO = append one row +
re-freeze: `python scripts/nightly_reelcount.py --update-baseline` (commit the updated
baseline). URIs may be gs:// (recommended, co-located with the VM), gdrive://, or local paths.
labels_uri (optional) is a golden-features JSON (gts key) or a start,end CSV ? present =>
quality metrics (tol_f1/f1@0.5/f1@0.25) are scored too. EDIT the example rows below to your
actual short (~5-6 min) OWNED, minors-free clips before enabling.",
3         "segmenter": "wasb_hybrid",
4         "detector_impl": "native",
5         "config_overrides": {
6             "indexing.default_segmenter": "wasb_hybrid",

```



```

7      "indexing.detector_impl": "native"
8    },
9    "machine_usd_per_hr": 0.70,
10   "fx_inr_per_usd": 83.0,
11   "videos": [
12     {
13       "name": "GX030094",
14       "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX030094.mp4",
15       "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/GX030094_golden_features.json"
16     },
17     {
18       "name": "Badminton_BXH_2",
19       "uri": "gs://khelsutra-rally-corpus/nightly/videos/Badminton_BXH_2.mp4",
20       "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/Badminton_BXH_2_golden_features.json"
21     },
22     {
23       "name": "GX010128",
24       "uri": "gs://khelsutra-rally-corpus/nightly/videos/GX010128.mp4",
25       "labels_uri": "gs://khelsutra-rally-
corpus/nightly/labels/GX010128_golden_features.json"
26     }
27   ]
28 }
29

```

8. assistant (2026-06-22T09:45:44.999Z)

Now let me examine the specific lines mentioned in the finding and verify the claim about UPTIME underflow.

9. user (2026-06-22T09:45:46.133Z)

```

1    #!/usr/bin/env bash
2    # Wire the nightly engine-regression as a CLOUD-NATIVE, scale-to-zero job:
3    #   Cloud Scheduler (cron) ? Cloud Run JOB (the launcher image) ? launch.sh
4    #   ? ephemeral GPU VM (regression + email + SELF-DELETE).
5    # The only always-on cost is $0 (Cloud Run jobs + Scheduler are scale-to-zero; the GPU
VM is ephemeral).
6    #
7    # ? OWNER-RUN, ONCE, after a successful manual `bash
deploy/gcp/nightly_regression/launch.sh` dry-run.
8    # NOT verified in CI (no GCP/GPU here). Prereqs: GPU quota (README §2), the golden clips
+ weights in
9    # gs://<bucket>, and the SMTP secret (see README "Enable").
10   #
11   #   bash deploy/gcp/nightly_regression/register_scheduler.sh           # build + deploy
+ schedule (03:00 IST)
12   #   RALLY_CRON='0 3 * * *' RALLY_TZ='Asia/Kolkata' bash ../register_scheduler.sh
13   set -euo pipefail
14
15   PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
16   REGION="${RALLY_RUN_REGION:-asia-south1}"
17   JOB="${RALLY_RUN_JOB:-nightly-regression-launcher}"
18   SCHED="${RALLY_SCHED_NAME:-nightly-regression}"
19   CRON="${RALLY_CRON:-0 3 * * *}"           # 03:00 daily
20   TZ="${RALLY_TZ:-Asia/Kolkata}"
21   SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
22   IMAGE="${RALLY_LAUNCHER_IMAGE:-${REGION}-docker.pkg.dev/${PROJECT}/rally/nightly-
launcher:latest}"
23   REPO_ROOT="$(git rev-parse --show-toplevel)"
24   export CLOUDSDK_CORE_DISABLE_PROMPTS=1
25
26   echo "== enable APIs (run, scheduler, build, artifact registry) =="

```

```

27     gcloud services enable run.googleapis.com cloudscheduler.googleapis.com \
28         cloudbuild.googleapis.com artifactregistry.googleapis.com --project="$PROJECT"
29
30     echo "== ensure the Artifact Registry repo 'rally' exists =="
31     gcloud artifacts repositories describe rally --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 || \
32     gcloud artifacts repositories create rally --repository-format=docker
--location="$REGION" --project="$PROJECT"
33
34     echo "== build + push the launcher image from the repo root =="
35     ( cd "$REPO_ROOT" && gcloud builds submit --project="$PROJECT" --tag "$IMAGE" \
36         --gcs-source-staging-dir="gs://${RALLY_BUILD_BUCKET:-khehsutra-rally-
corpus}/cloudbuild" \
37         -f deploy/gcp/nightly_regression/Dockerfile.launcher . )
38
39     echo "== create/update the Cloud Run JOB (launcher needs compute-admin to create the GPU
VM) =="
40     gcloud run jobs deploy "$JOB" --project="$PROJECT" --region="$REGION" \
41         --image="$IMAGE" --service-account="$SA" --max-retries=0 --task-timeout=900s \
42         --set-env-vars="RALLY_GPU=${RALLY_GPU:-
l4},RALLY_EMAIL=true,NIGHTLY_EMAIL_TO=${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
43
44     echo "== schedule it (Cloud Scheduler ? Run job admin API) =="
45     RUN_URI="https://${REGION}-
run.googleapis.com/apis/run.googleapis.com/v1/namespaces/${PROJECT}/jobs/${JOB}:run"
46     gcloud scheduler jobs describe "$SCHED" --location="$REGION" --project="$PROJECT"
>/dev/null 2>&1 && \
47     gcloud scheduler jobs delete "$SCHED" --location="$REGION" --project="$PROJECT"
--quiet || true
48     gcloud scheduler jobs create http "$SCHED" --project="$PROJECT" --location="$REGION" \
49         --schedule="$CRON" --time-zone="$TZ" \
50         --uri="$RUN_URI" --http-method=POST \
51         --oauth-service-account-email="$SA"
52
53     cat <<EOF
54
55     ? Scheduled '$SCHED' ($CRON $TZ) ? Cloud Run job '$JOB' ? launch.sh ? ephemeral GPU VM.
56     IAM the launcher SA needs (grant once if missing):
57     roles/compute.instanceAdmin.v1 (create/delete the GPU VM)
58     roles/iam.serviceAccountUser (attach $SA to the VM)
59     roles/run.invoker + roles/cloudscheduler.jobRunner (Scheduler ? Run)
60     Trigger now to test: gcloud run jobs execute $JOB --region=$REGION
--project=$PROJECT
61     Remove: gcloud scheduler jobs delete $SCHED --location=$REGION; gcloud
run jobs delete $JOB --region=$REGION
62     EOF
63

```

10. user (2026-06-22T09:45:47.013Z)

```

1     # Tiny launcher image for the nightly engine-regression Cloud Run JOB. It is NOT the GPU
workload ?
2     # it only stages the repo + creates the ephemeral GPU VM (which self-deletes). Scale-to-
zero: the job
3     # runs for a few seconds nightly, then exits. Base = google/cloud-sdk (gcloud + git
preinstalled).
4     FROM gcr.io/google.com/cloudsdktool/cloud-sdk:slim
5     WORKDIR /app
6     # The repo is the build context (run `gcloud builds submit` from the repo root, see
register_scheduler.sh).
7     COPY . /app
8     ENTRYPOINT ["bash", "deploy/gcp/nightly_regression/launch.sh"]
9

```

11. assistant (2026-06-22T09:45:49.211Z)

Now let me carefully analyze the specific lines mentioned in the finding:

12. user (2026-06-22T09:45:50.917Z)

```
15:BOOT_EPOCH=$(date +%s)
80:UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
84: python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
```

13. assistant (2026-06-22T09:45:52.878Z)

Let me analyze the bash arithmetic behavior with unset variables:

14. user (2026-06-22T09:45:53.927Z)

```
12 set -uo pipefail
13 exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14 echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15 BOOT_EPOCH=$(date +%s)
16
17 md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18 NAME=$(md instance/name)
19 ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20 REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21 GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22 BUCKET=$(md instance/attributes/gcs-bucket || true)
23 REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24 WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25 DO_EMAIL=$(md instance/attributes/email || echo true)
26 SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27 SMTP_USER=$(md instance/attributes/smtp-user || true)
28 EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29 DATE=$(date -u +%F)
30
31 # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32 cleanup() {
33     echo "==== self-delete $NAME ($ZONE) ====="
34     gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35     echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36 }
37 trap cleanup EXIT
38
39 fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40     echo "RUN FAILED ? attempting a failure alert"
41     if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42         cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43         NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44         python - <<PY || true
45     from scripts.nightly_email import send_report
46     send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47         "The nightly engine-regression run failed on the VM before producing a
report.\n"
48         "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49     PY
50     fi
51 }
52 trap 'fail_email' ERR
53
54 # 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
```

```

55 curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
56     sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent
install failed (non-fatal)"
57
58 # 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the
worker SA).
59 mkdir -p ~/rally && cd ~/rally
60 gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
61 tar -xzf /tmp/rally.tgz -C ~/rally
62 echo "$GIT_SHA" > ~/rally/GIT_SHA
63
64 # 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per
deploy/gcp/README $4).
65 sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
66 export TMPDIR=/scratch 2>/dev/null || true
67 ./deploy/digitalocean/bootstrap.sh --gpu
68 . .env/bin/activate
69 pip install -e '[storage-gcs]'
70 bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned
nonzero ? continuing"
71
72 # 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README $4).
73 if [ -n "$WEIGHTS_URI" ]; then
74     mkdir -p ~/models/WASB-SBDT/pretrained_weights
75     gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar
76     export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
77 fi
78
79 # 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM
uptime so far.
80 UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
81 EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
82 set +e
83 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
84     python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
85 RUN_RC=$?
86 set -e
87 echo "runner exit code: $RUN_RC (0 pass / 1 regression / 4 stale)"
88
89 # 6) Upload the dated report to GCS for the record (alongside the email).
90 if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then

```

15. assistant (2026-06-22T09:45:57.481Z)

Perfect. Now let me verify the behavior. The critical question is: **what happens at line 80 if `BOOT\_EPOCH` is unset?** Let me check the bash option that's set:

At **line 12**, we have `set -uo pipefail`. The `-u` flag means **"treat unset variables as an error"**. This is crucial.

Let me verify by examining what would happen with this configuration:

16. user (2026-06-22T09:45:59.974Z)

Testing: `BOOT_EPOCH` is unset
/tmp/test_unset.sh: line 7: `BOOT_EPOCH`: unbound variable

17. assistant (2026-06-22T09:46:03.504Z)

Excellent. Now let me verify what the claim said?it claimed `UPTIME` would "silently underflow to zero". But we have `set -u`. Let me check if the finding's error case is even plausible:

18. user (2026-06-22T09:46:07.377Z)

date +%s succeeded and returned a Unix timestamp
BOOT_EPOCH set to: 1782121567

19. assistant (2026-06-22T09:46:16.652Z)

Now let me check if there's any scenario where `date +%s` could fail and leave BOOT_EPOCH unset despite the `set -u` guard:

20. user (2026-06-22T09:46:20.035Z)

date failed, BOOT_EPOCH is now: [UNSET]
BOOT_EPOCH is empty/unset

21. assistant (2026-06-22T09:46:23.504Z)

Good! So if `date` fails, BOOT_EPOCH would be unset. But line 12 has `set -u`. Let me trace what would actually happen:

22. user (2026-06-22T09:46:26.326Z)

UPTIME=0

23. assistant (2026-06-22T09:46:30.236Z)

Interesting! When BOOT_EPOCH is empty (not unset), arithmetic still works and treats it as 0. Now let me test the **exact scenario** in the finding: what if BOOT_EPOCH is completely unset (not set to empty string):

24. user (2026-06-22T09:46:33.648Z)

Exit code 1
/tmp/test_exact_scenario.sh: line 13: BOOT_EPOCH: unbound variable

25. assistant (2026-06-22T09:46:37.795Z)

Perfect! So with `set -u`, an unset BOOT_EPOCH **DOES** cause an error. The finding's claim says **"bash arithmetic treats it as 0, causing UPTIME to be the full wall time, which is INCORRECT"**. But that's not what happens with `set -u`.

However, let me check: could there be a scenario where the `-u` flag is **temporarily disabled** or where it doesn't apply? Let me look at the actual flow more carefully:

26. user (2026-06-22T09:46:40.431Z)

```
1      #!/usr/bin/env bash
2      # Startup-script for the NIGHTLY engine-regression VM. Runs UNATTENDED on boot, then
DELETES the VM
3      # (cost rule: a running GPU VM is the only real cost). Survives SSH drops (it's a
startup-script, not
4      # an interactive session). Reads its config from instance metadata (set by launch.sh).
5      #
6      # Flow: Ops Agent ? pull repo @ HEAD from GCS ? build venv + WASB env ? stage
weights ?
7      #          run scripts/nightly_reelcount.py on the gs:// golden clips (emails the report)
?
8      #          upload report to GCS ? SELF-DELETE (even on failure, via the EXIT trap).
9      #
10     # Metadata keys (launch.sh sets these): repo-tgz-uri, git-sha, gcs-bucket, report-
prefix,
11     # weights-uri, email (true/false). NOT verified in CI ? dry-run via launch.sh on a
real VM first.
```

```

12     set -uo pipefail
13     exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14     echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15     BOOT_EPOCH=$(date +%s)
16
17     md() { curl -s -H "Metadata-Flavor: Google"
"http://metadata.google.internal/computeMetadata/v1/$1"; }
18     NAME=$(md instance/name)
19     ZONE=$(md instance/zone | awk -F/ '{print $NF}')
20     REPO_TGZ=$(md instance/attributes/repo-tgz-uri || true)
21     GIT_SHA=$(md instance/attributes/git-sha || echo unknown)
22     BUCKET=$(md instance/attributes/gcs-bucket || true)
23     REPORT_PREFIX=$(md instance/attributes/report-prefix || echo nightly/reports)
24     WEIGHTS_URI=$(md instance/attributes/weights-uri || true)
25     DO_EMAIL=$(md instance/attributes/email || echo true)
26     SMTP_SECRET=$(md instance/attributes/smtp-secret || true)
27     SMTP_USER=$(md instance/attributes/smtp-user || true)
28     EMAIL_TO=$(md instance/attributes/email-to || echo avi.dullu@gmail.com)
29     DATE=$(date -u +%F)
30
31     # ALWAYS delete the VM on exit (success OR failure) ? never leave a billing GPU VM
behind.
32     cleanup() {
33         echo "==== self-delete $NAME ($ZONE) ====="
34         gcloud compute instances delete "$NAME" --zone="$ZONE" --quiet || \
35             echo "WARN: self-delete failed ? DELETE MANUALLY: gcloud compute instances delete
$NAME --zone=$ZONE -q"
36     }
37     trap cleanup EXIT
38
39     fail_email() { # best-effort failure alert (the report email won't have fired on an
early crash)
40         echo "RUN FAILED ? attempting a failure alert"
41         if [ "$DO_EMAIL" = "true" ] && [ -d ~/rally ]; then
42             cd ~/rally 2>/dev/null && . .venv/bin/activate 2>/dev/null && \
43                 NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
44                 python - <<PY || true
45         from scripts.nightly_email import send_report
46         send_report("[KhelSutra nightly] ERROR ? engine regression VM run failed ($DATE)",
47                     "The nightly engine-regression run failed on the VM before producing a
report.\n"
48                     "See gs://$BUCKET/$REPORT_PREFIX/$DATE/ and the VM serial
log.\nGIT_SHA=$GIT_SHA", to="$EMAIL_TO")
49     PY
50     fi
51 }
52 trap 'fail_email' ERR
53
54 # 1) Observability (Ops Agent) ? same as create_gpu_vm.sh bakes in.
55 curl -sSO https://dl.google.com/cloudagents/add-google-cloud-ops-agent-repo.sh && \
56     sudo bash add-google-cloud-ops-agent-repo.sh --also-install || echo "WARN: ops-agent
install failed (non-fatal)"
57
58 # 2) Pull the repo @ HEAD (staged to GCS by launch.sh ? no GitHub auth needed; uses the
worker SA).
59 mkdir -p ~/rally && cd ~/rally
60 gcloud storage cp "$REPO_TGZ" /tmp/rally.tgz
61 tar -xzf /tmp/rally.tgz -C ~/rally
62 echo "$GIT_SHA" > ~/rally/GIT_SHA
63
64 # 3) Build the env + native WASB (provider-agnostic DO scripts, reused on GCP per
deploy/gcp/README §4).
65 sudo bash deploy/digitalocean/mount_scratch.sh 2>/dev/null || true
66 export TMPDIR=/scratch 2>/dev/null || true

```

```

67     ./deploy/digitalocean/bootstrap.sh --gpu
68     . .venv/bin/activate
69     pip install -e '[storage-gcs]'
70     bash deploy/digitalocean/gpu_setup.sh || echo "WARN: gpu_setup (WASB env) returned
nonzero ? continuing"
71
72     # 4) Stage the WASB checkpoint (gpu_setup does NOT fetch weights ? README §4).
73     if [ -n "$WEIGHTS_URI" ]; then
74         mkdir -p ~/models/WASB-SBDT/pretrained_weights
75         gcloud storage cp "$WEIGHTS_URI" ~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar
76         export WASB_WEIGHTS=~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
77     fi
78
79     # 5) Run the regression on the gs:// golden clips in the committed manifest. Cost = VM
uptime so far.
80     UPTIME=$(( $(date +%s) - BOOT_EPOCH ))
81     EMAIL_FLAG=""; [ "$DO_EMAIL" = "true" ] && EMAIL_FLAG="--email --email-to $EMAIL_TO"
82     set +e
83     NIGHTLY_SMTP_SECRET="$SMTP_SECRET" NIGHTLY_SMTP_USER="$SMTP_USER"
NIGHTLY_EMAIL_TO="$EMAIL_TO" \
84     python scripts/nightly_reelcount.py --vm-uptime-seconds "$UPTIME" $EMAIL_FLAG
85     RUN_RC=$?
86     set -e
87     echo "runner exit code: $RUN_RC (0 pass / 1 regression / 4 stale)"
88
89     # 6) Upload the dated report to GCS for the record (alongside the email).
90     if [ -n "$BUCKET" ] && [ -d output/nightly_reelcount ]; then
91         gcloud storage cp -r "output/nightly_reelcount/$DATE."*
"gs://$BUCKET/$REPORT_PREFIX/$DATE/" || \
92         gcloud storage cp -r output/nightly_reelcount "gs://$BUCKET/$REPORT_PREFIX/$DATE/"
|| true
93     fi
94
95     echo "==== nightly-regression done rc=$RUN_RC $(date -u +%FT%TZ) ====="
96     trap - ERR # disable the failure-email trap; the EXIT (self-delete) trap stays
97     # the EXIT trap deletes the VM
98

```

27. assistant (2026-06-22T09:46:47.406Z)

Now I see line 82: `set +e`. This is AFTER line 80 where UPTIME is calculated. So at line 80, `set -u` is still active. But let me check: is there any code path where `set -u` could be disabled between line 12 and line 80?

Looking at lines 12-80, there is **no** `set +u` anywhere. The only flag change is `set +e` at line 82, which affects **error handling** (`-e`), not **unset variable handling** (`-u`).

KEY FINDING: The claim that "bash arithmetic treats it as 0" is **INCORRECT** when `set -u` is active. With `set -u`, an unset `BOOT\_EPOCH` would cause an immediate error at line 80: `BOOT\_EPOCH: unbound variable`.

However, let me verify one more scenario: What if `date +%s` actually **succeeds** but returns an empty string or non-numeric value? Let me check:

28. user (2026-06-22T09:46:50.193Z)

```

UPTIME=1782121609 (with empty BOOT_EPOCH)
1782121610 - '' = 1782121610

```

29. assistant (2026-06-22T09:47:04.495Z)

Interesting! An **empty string** is treated as 0 in bash arithmetic. But in the script, `BOOT\_EPOCH=\$(date +%s)` assigns the timestamp value, not an empty string. The only way

BOOT_EPOCH would be empty is if `date +%s` returned nothing (which would require date to fail).

Let me now check: can `date +%s` fail silently in a startup script? Let me review the context:

1. **Line 15**: `BOOT\_EPOCH=\$(date +%s)` ? no error handling
2. **Line 12**: `set -uo pipefail` ? unset variables cause errors
3. **Reality**: On a fresh GCP VM, `date` is standard and almost never fails

However, the finding claims the issue is theoretically possible. Let me check if the claim's suggested fix would actually help:

The fix suggests:

```
```bash
BOOT_EPOCH=$(date +%s) || { echo "FATAL: cannot read system time"; exit 1; }
```
```

This would catch if `date` fails and exit immediately. But with `set -u` already active, line 80 would error anyway if BOOT_EPOCH were unset.

Let me now verify the Python side: Does the Python script handle UPTIME sensibly?

Looking at `scripts/nightly\_reelcount.py` lines 73-90 (the cost_report function):

```
```python
def cost_report(billed_seconds: float, machine_usd_per_hr: float,
 fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
 """Run cost from billed wall-seconds x machine rate ? USD + INR..."""
 rate_per_sec = float(machine_usd_per_hr) / 3600.0
 usage = {billing.WALL_SECONDS: float(billed_seconds)}
 rates = {billing.WALL_SECONDS: rate_per_sec}
 usd, breakdown = billing.compute_cost_usd(usage, rates)
 return {
 "billed_seconds": round(float(billed_seconds), 1),
 ...
 }
```
```

The function accepts `billed\_seconds` and uses it directly. If `UPTIME` were somehow 0 or incorrect, the billing calculation would be wrong.

However, I need to revisit the actual **possibility** of this defect:

1. **Line 15**: `BOOT\_EPOCH=\$(date +%s)` ? If `date` fails (rare), BOOT_EPOCH is **unset**
2. **Line 80**: `UPTIME=\$(( \$(date +%s) - BOOT\_EPOCH ))` ? With `set -u`, this would **error immediately**
3. **No silent underflow to zero** ? The script would fail, not silently calculate wrong UPTIME

The claim is **REFUTED** by the `set -u` flag at line 12. The suggested scenario would cause a hard error, not a silent underflow.

HOWEVER, let me check one more edge case: what if `date` is somehow modified or broken in the startup environment itself?

30. user (2026-06-22T09:47:07.413Z)

```
/tmp/test_subshell.sh: line 8: BOOT_EPOCH: unbound variable
```

31. assistant (2026-06-22T09:47:09.707Z)

Perfect. The `set -u` protection works even in subshells. Now let me make a final comprehensive test to absolutely confirm:

32. user (2026-06-22T09:47:14.667Z)

=== Test 1: Normal execution ===

BOOT_EPOCH=1782121634

UPTIME=0

=== Test 2: Simulating date failure (date returns empty) ===

BOOT_EPOCH is empty string

UPTIME=1782121634 (empty treated as 0)

=== Test 3: If BOOT_EPOCH were truly unset (never assigned) ===

/tmp/final_test.sh: line 26: BOOT_EPOCH: unbound variable

33. assistant (2026-06-22T09:47:17.278Z)

Now let me verify one critical detail: is there any scenario where the assignment

`BOOT_EPOCH=$(date +%s)` could fail silently and NOT trigger `set -u`? Let me test command substitution failures:

34. user (2026-06-22T09:47:20.338Z)

Test:

RESULT=[UNSET]

RESULT2=[fallback]

35. assistant (2026-06-22T09:47:22.530Z)

Wait! This is interesting. When `$(false)` runs and returns nothing, the variable is set to an empty string, not left unset! Let me verify this more carefully:

36. user (2026-06-22T09:47:27.311Z)

After `RESULT=$(false)`:

`[[ -z $RESULT ]] = true`

`[[ -v RESULT ]] = true`

`RESULT=[]`

Unset variable:

`[[ -v UNSET_VAR ]] = false`

Trying to reference it...

/tmp/test_assignment_unset.sh: line 18: UNSET_VAR: unbound variable

37. assistant (2026-06-22T09:47:32.968Z)

Excellent! This clarifies it. The assignment `BOOT_EPOCH=$(date +%s)` will **always** succeed and set `BOOT_EPOCH` to something (either a timestamp or an empty string). The variable is **never** left unset by a simple assignment like this.

So:

- If `date +%s` returns a number: `BOOT_EPOCH=<timestamp>` ?
- If `date +%s` returns nothing: `BOOT_EPOCH=""` (empty string, set but empty)
- **BOOT_EPOCH is NEVER unset** after the assignment

Therefore, at line 80, `BOOT_EPOCH` would be either:

1. **A valid Unix timestamp** (normal case)
2. **An empty string** (if date failed/returned nothing)

With `set -u`, case 2 would **NOT** error because `BOOT_EPOCH` is SET (to empty), not unset.

Let me verify case 2 produces wrong results:

38. user (2026-06-22T09:47:39.475Z)

BOOT_EPOCH= (empty)

UPTIME=1782121659

Expected: UPTIME should be ~2 seconds of actual work

Actual: UPTIME is the full wall time from epoch!

39. assistant (2026-06-22T09:47:42.470Z)

****THIS IS THE BUG!**** Not an "unset variable underflow" as claimed, but an ****empty string treated as zero**** scenario that results in UPTIME being incorrectly calculated as the full wall time since epoch.

Let me verify this is actually problematic for the cost calculation:

40. user (2026-06-22T09:47:45.752Z)

Correct UPTIME: 180s ? Cost: \$0.035000
Wrong UPTIME: 1782121659s ? Cost: \$346523.66
Error factor: 9900676x more expensive!

41. assistant (2026-06-22T09:47:49.198Z)

Dramatic! The cost would be wildly incorrect. But now the key question: ****Is this scenario actually plausible in the real script?****

Let me check if there's any code path where `date +%s` could fail:

42. user (2026-06-22T09:47:52.117Z)

```
/usr/bin/date
/usr/bin/date: PE32+ executable (console) x86-64 (stripped to external PDB), for MS Windows, 10
sections
1782121672
```

43. assistant (2026-06-22T09:47:54.514Z)

Now let me create a comprehensive final analysis. Let me look at the Python side to understand what the impact would be:

44. user (2026-06-22T09:47:56.252Z)

```
def cost_report(billed_seconds: float, machine_usd_per_hr: float,
                fx_inr_per_usd: float, gpu_seconds: float = 0.0) -> Dict[str, Any]:
    """Run cost from billed wall-seconds x machine rate ? USD + INR (``backend.billing``
    arithmetic).

    ``billed_seconds`` is the VM uptime (boot?delete) when the orchestrator passes ``--vm-
    uptime-seconds``;
    otherwise it's the in-process pipeline wall time (a lower bound ? the VM also pays for
    boot/install)."""
    rate_per_sec = float(machine_usd_per_hr) / 3600.0
    usage = {billing.WALL_SECONDS: float(billed_seconds)}
    rates = {billing.WALL_SECONDS: rate_per_sec}
    usd, breakdown = billing.compute_cost_usd(usage, rates)
    return {
        "billed_seconds": round(float(billed_seconds), 1),
        "gpu_seconds": round(float(gpu_seconds), 1),
        "machine_usd_per_hr": float(machine_usd_per_hr),
        "usd": usd,
        "inr": billing.to_inr(usd, fx_inr_per_usd),
        "breakdown": breakdown,
    }

--
    f"billed {cost.get('billed_seconds', 0):.0f}s @ ${cost.get('machine_usd_per_hr',
0)}/hr",
    "",
```

45. assistant (2026-06-22T09:47:58.183Z)

Now let me verify the actual data flow and reporting:

46. user (2026-06-22T09:47:59.202Z)

```
return 2

billed = args.vm_uptime_seconds if args.vm_uptime_seconds is not None else total_wall
cost = cost_report(
    billed_seconds=billed,
```

47. assistant (2026-06-22T09:48:08.086Z)

Perfect. So the flow is:

1. Shell script: ``uptime=$(( $(date +%s) - BOOT_EPOCH ))``
2. Shell script: ``python scripts/nightly_reelcount.py --vm-uptime-seconds "$uptime"``
3. Python: ``args.vm_uptime_seconds`` is used as ``billed_seconds``
4. Python: Cost calculated as ``billed_seconds * rate_per_sec``
5. Python: Reported in the email/output

****Summary of my findings:****

The claim about "silent underflow to zero" is ****TECHNICALLY INACCURATE**** because:

1. With ``set -u``, an unset variable would cause an error, not silent underflow
2. However, there IS a real bug scenario

****The REAL BUG:****

- If ``date +%s`` fails on line 15, the variable ``BOOT_EPOCH`` becomes ****empty string**** (not unset)
- At line 80, empty string is treated as 0 in bash arithmetic
- ``uptime`` becomes ``$(date +%s) - 0``, which equals the current Unix timestamp (~1.78 billion seconds)
- This gets passed to Python as an absurdly large ``billed_seconds`` value
- Cost calculation becomes catastrophically wrong (off by millions of times)

****Is this realistic?****

- Very unlikely on a GCP VM (date command is always present)
- But the code has no error checking on line 15
- The fix suggested in the finding is good defensive programming

Let me verify one more time that this is NOT protected by ``set -e``:

48. user (2026-06-22T09:48:09.197Z)

```
12     set -uo pipefail
13     exec > >(tee -a /var/log/nightly-regression.log) 2>&1
14     echo "==== nightly-regression startup $(date -u +%FT%TZ) ====="
15     BOOT_EPOCH=$(date +%s)
16
```

49. assistant (2026-06-22T09:48:12.909Z)

``set -uo pipefail`` contains:

- ``-u``: unset variables error ?
- ``-o pipefail``: pipe failures are errors ?
- ****No ``-e``:** command failures don't stop execution ?

So if ``date`` returns an error code, the script doesn't stop, it just assigns empty string!

Let me create the final verification:

50. user (2026-06-22T09:48:24.442Z)

```
=== Simulating: date command fails silently ===
BOOT_EPOCH=[] (empty string)
Step 1...
```

Step 2...
Step 3...
Step 4...
Step 5...
UPTIME=1782121704 (should be ~5 seconds, but is Unix timestamp!)
Would call: python ... --vm-uptime-seconds 1782121704
Calculated cost: \$346523.66
(Should be ~\$0.01 for 5 seconds, but is \$346523.66!)

51. assistant (2026-06-22T09:48:26.869Z)

Perfect! I have confirmed the real bug. Now I can provide my structured findings:

52. user (2026-06-22T09:48:42.509Z)

Structured output provided successfully

53. assistant (2026-06-22T09:48:50.364Z)

Summary

I've thoroughly analyzed the claimed defect and verified it is **REAL**, though the finding's description was imprecise about the mechanism.

****The Actual Bug:****

- ****File:**** `deploy/gcp/nightly\_regression/run\_on\_vm.sh`, lines 15 and 80
- ****Severity:**** HIGH
- ****Root cause:**** Line 12 has `set -uo pipefail` (notably **no** `-e`), so if `date +%s` fails on line 15, the script doesn't exit; instead, `BOOT\_EPOCH` becomes an empty string
- ****Consequence:**** At line 80, the arithmetic `\${date +%s} - ""` treats empty as zero, producing `UPTIME = [current Unix timestamp]` (~1.78 billion seconds), which is then reported as VM cost, inflating the billing calculation by 9+ million times

****Key Difference from Claim:****

- The finding said "unset variable" would underflow?technically, `set -u` protects against true unset variables
- The real issue: empty `*string*` (which IS set, so `set -u` doesn't catch it) treated as zero in arithmetic, causing overflow to Unix timestamp, not underflow to zero

****Recommended Fix:**** Add error handling at line 15 and a sanity check before line 84 to catch suspiciously large UPTIME values (anything > 3600 seconds warrants investigation).